# Waseem Raza, PhD

## Curriculum Vitae | AI/ML for Wireless Networks & Intelligent Systems

Electrical & Computer Engineering | AI/ML for Wireless Networks | 5G/6G | Reliable Learning | Engineering Education

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LinkedIn | GitHub | Google Scholar | Portfolio

## Academic Profile

Electrical and Computer Engineering researcher, educator, and machine-learning engineer with a PhD from the University of Oklahoma. Doctoral and continuing research focuses on AI/ML for wireless networks, including resilient and data-efficient learning, 5G/6G network intelligence, propagation modeling and digital twins, optimization, self-healing, transfer learning, domain adaptation, generative modeling, reinforcement learning, and experimental wireless systems. Academic experience spans university teaching, research mentoring, laboratory instruction, curriculum and accreditation support, publication development, and interdisciplinary research, complemented by industrial R&D at AT&T Labs, Futurewei, Impac, and AT&T.

### Research Interests

AI/ML for Wireless and 5G/6G Networks | Reliable, Robust, and Data-Efficient Learning | Network Optimization, Self-Healing, and Autonomous Systems | Propagation Modeling, Digital Twins, and Synthetic Data | Transfer Learning, Domain Adaptation, and Generative Modeling | Reinforcement Learning and Sequential Decision-Making | Wearable Intelligence and Human-Centered Sensing

### Doctoral Research Focus

Artificial Intelligence and Machine Learning for Wireless Networks, with emphasis on resilient learning, data-efficient modeling, network optimization, propagation and digital-twin modeling, self-healing cellular networks, and reliable AI under real-world variability.

## Education

**PhD, Electrical and Computer Engineering** *09/2019 - 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

**MSc, Telecommunication Engineering** *08/2014 - 08/2016*
University of Engineering and Technology, Taxila, Pakistan

**BSc, Telecommunication Engineering** *11/2010 - 06/2014*
University of Engineering and Technology, Taxila, Pakistan

### Relevant Graduate Coursework

Telecommunication Technology / LTE Networks | Emerging Topics in LTE-Advanced and 5G | Advanced Machine Learning / Computer Vision | Testbed Design for Emerging Cellular Networks | Numerical Analysis | Information Theory and Probability | Digital Signal Processing

## Teaching, Research & Professional Experience

### Teaching & Instruction

**Lecturer, Department of Technology** **12/2017 - 09/2019**
*University of Lahore, Pakistan* *Lahore, Pakistan*

- Taught undergraduate Electrical Engineering Technology courses including Computer Programming, Communication Systems, and Wireless Communication, connecting theory with implementation-oriented examples.
- Mentored and advised students on academic projects, technical problem solving, and career pathways, including senior-design and course-project development.
- Contributed to curriculum development and course-content alignment with program outcomes, industry needs, and accreditation requirements.
- Supported departmental accreditation documentation, examinations, student orientation, technical workshops, seminars, and academic administration.

**Teaching areas:** Wireless Communications | Digital and Analog Communication Systems | 5G/6G and Emerging Wireless Networks | Machine Learning and Artificial Intelligence | Signals and Systems | Digital Signal Processing | Computer Programming | Data Science and Experimental Methods

**Student mentoring and research development**

- Mentored undergraduate and graduate researchers on machine-learning implementation, wireless-network experimentation, simulation, data analysis, and technical writing.
- Supported Bachelor's and Master's thesis/project work in communications and wireless systems, including research methodology, experiment planning, manuscript preparation, and presentation.
- Advised students on senior/capstone projects, academic planning, and technical career development.

- Developed assignments, laboratory material, and implementation-oriented exercises connecting communication theory with practical engineering systems.

## University Research Appointments

**Graduate Research Assistant** **09/2019 - 07/2025**
*AI4Networks, University of Oklahoma* *Tulsa, OK*

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

**Research Associate, ComSens Lab** **06/2016 - 12/2017**
*COMSATS University Islamabad, Pakistan* *Islamabad, Pakistan*

- Conducted research in wireless communication systems and contributed to publication-oriented modeling, simulation, analysis, and technical writing.
- Mentored Bachelor's and Master's students on thesis projects, research methodology, experiment design, technical writing, and presentation of results.
- Reviewed research manuscripts and student technical reports, providing structured feedback aligned with publication and academic-quality standards.
- Assisted with research-proposal preparation, literature synthesis, lab seminars, paper-review meetings, and coordination of research activities.

**Full-Time Research Scholar, Communication Systems Lab** **08/2014 - 02/2016**
*University of Engineering and Technology, Taxila, Pakistan* *Taxila, Pakistan*

- Conducted full-time graduate research in communication systems and wireless engineering while completing the MSc in Telecommunication Engineering.
- Assisted faculty with Communication Systems and Digital Communication Systems instruction, student questions, assignments, grading, and exam preparation.
- Supervised laboratory sessions for Communication Systems and Wireless Communication and developed lab manuals and implementation-oriented exercises.
- Supported curriculum development and student technical-project activities connecting theoretical communication concepts with practical experimentation.

## Industry Research & Engineering

**Machine Learning Engineer** **02/2025 - Present**
*AT&T, contractor via Innovatix Inc.* *Bedminster, NJ*

- Translate machine-learning research into operational telecommunications systems, emphasizing domain adaptation, reliability, model validation, and responsible decision support.
- Develop reusable cross-domain evaluation and error-analysis methods for AI systems operating under distribution shift and heterogeneous operational conditions.
- Collaborate across engineering, software, and public-safety teams to connect research methods with real-world network-assurance requirements.

**Data Scientist and AI/ML Engineer** **09/2024 - 12/2024**
*Impac Exploration Services* *Weatherford, OK*

- Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data.
- Developed scalable ingestion, cleansing, transformation, and validation workflows using Python, Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.
- Collaborated with geologists to convert domain requirements and technical information into AI-ready datasets and actionable analytical outputs.

**Advanced-Degrees Intern** **06/2024 - 08/2024**
*AT&T Labs* *Austin, TX*

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

**Advanced Wireless Technology Development Intern** **09/2022 - 01/2023**
*Futurewei Technologies* *Schaumburg, IL*

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

**Advanced-Degrees Intern** **06/2022 - 08/2022**
*AT&T Labs* *Bedminster, NJ*

- Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%.
- Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.
- Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.

## Research Contributions

### Publications

**Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches** *2026*
*HealthINF | Accepted*

**AI-Powered Resilience: A Dual-Approach for Outage Management in Dense Cellular Networks** *2025*
*Computer Communications*

**On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks** *2025*
*Sensors 2025*

**An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin** *2024*
*IEEE PIMRC*

**Refining Wireless Propagation Models using Domain-Informed GANs amid Data Scarcity** *2024*
*IEEE VTC-Fall 2024*

**Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks** *2024*
*IEEE SmartNets*

**An AI-Enabled Framework to Defend Ingenious MDT-Based Attacks on the Emerging Zero Touch Cellular Networks** *2024*
*IEEE Network, 38(1), 228-237*

**Positioning Error Impact Compensation through Data-Driven Optimization in User-Centric Networks** *2023*
*IEEE GLOBECOM 2023*

**Toward a Hybrid RF/Optical Lunar Communication System (LunarComm)** *2022*
*IEEE Network*

**Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN** *2022*
*IEEE WCNC*

**Unification of RF Energy Harvesting Schemes under Mixed Rayleigh-Rician Fading Channels** *2020*
*AEÜ - International Journal of Electronics and Communications*

### Research Projects & Platforms

**SyntheticNET: 3GPP-Compliant Network Simulation** *Research Project | 2020-2024*

Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.

*Tools & Methods:* Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

**TurboRAN Experimental Platform** *Research Infrastructure | 2020-2024*

Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled network conditions. Supported repeatable measurement collection, testbed experimentation, RAN optimization studies, and validation of emerging wireless-network algorithms.

*Tools & Methods:* 5G Testbeds, RAN, Network Measurement, Experiment Design, Wireless Prototyping, Multi-Band Systems, Performance Evaluation, RAN Optimization

**Propagation Models & Digital Twins** *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

*Tools & Methods:* Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

**Wearable Intelligence for Health Monitoring** *Research Project | 2024-2026*

Developed deep-learning workflows for recognizing medication-taking and dietary activities from smartwatch accelerometer and gyroscope signals in free-living settings. Evaluated windowing, feature extraction, CNN/LSTM-style sequence models, cross-user generalization, and recall-oriented performance for health-monitoring applications.

*Tools & Methods:* Wearable AI, Smartwatch IMU, Time Series, Deep Learning, CNN/LSTM, Feature Engineering, Cross-User Evaluation, Activity Recognition, Healthcare AI

**AI for Telecom Network & Security Optimization** *Certification Project | 2026*

Applied AI concepts to telecommunications network optimization, security analysis, and customer-experience improvement through a structured specialization project. Framed telecom use cases around operational data, AI-assisted analysis, optimization opportunities, and practical service-improvement recommendations.

*Tools & Methods:* AI for Telecommunications, Network Optimization, Telecom Security, Customer Experience, Python, TensorFlow, Operational Analytics, Service Improvement

**Prompt Engineering & Foundation Model Workflows** *Certification Project | 2026*

Applied prompt-engineering patterns, context design, model/platform comparison, and responsible-AI considerations across practical generative-AI exercises. Iteratively refined prompts and task workflows to improve relevance, structure, and reliability of foundation-model outputs.

*Tools & Methods:* Prompt Engineering, Foundation Models, Context Design, IBM watsonx, Hugging Face, Responsible AI, Output Evaluation, Generative AI

## Research Methods & Technical Expertise

**AI/ML Methods:** Deep learning, transfer learning, domain adaptation, adversarial learning, GANs, autoencoders, reinforcement learning, classification, regression, anomaly detection, time-series modeling, and model evaluation

**Wireless Systems:** LTE/4G, 5G-NR, NTN, RAN, mobility, propagation, positioning, outage management, self-healing, network measurements, digital twins, and service assurance

**Simulation & Experimentation:** 3GPP-compliant simulation, synthetic data, experimental testbeds, statistical analysis, robustness testing, cross-validation, distribution-shift evaluation, and reproducible research workflows

**Programming & Data:** Python, SQL, PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Pandas, NumPy, SciPy, visualization, cloud analytics, Spark/Databricks, Git, and Linux

**Teaching & Academic Practice:** Communication systems, wireless communications, programming, laboratory instruction, research methods, technical writing, student mentoring, curriculum development, and academic service

## Academic Service & Recognition

### Academic Service

- Contributed to curriculum development and course-content alignment with engineering program and industry needs.
- Supported departmental accreditation and quality-assurance documentation, examination processes, and academic administration.
- Reviewed research manuscripts and student technical reports and provided publication-oriented feedback.
- Participated in research seminars, paper-review meetings, student orientations, technical workshops, and academic events.

- Assisted research teams with proposal preparation, literature synthesis, experimental planning, and documentation.

### Honors & Awards

**William H. Barkow Scholarship** *2024*
University of Oklahoma, Electrical and Computer Engineering

**Clyde L. Farrar Endowment Scholarship** *2024*
University of Oklahoma

**P.H. Robinson Fellowship** *2023*
University of Oklahoma

**FAS-Charles Forgan Scholarship** *2023*
University of Oklahoma

**Nettie Vincent Boggs Graduate Endowment** *2022*
University of Oklahoma

**SHS-Paul D. Massad Scholarship** *2022*
University of Oklahoma

**Cleo Cross Memorial Scholarship** *2021*
University of Oklahoma

**Marjorie Haskell Engineering Scholarship** *2021*
University of Oklahoma

**Fully Funded PhD / Graduate Research Assistantship** *2019*
University of Oklahoma

## Professional Development

1. **Machine Learning** — Stanford Online / Coursera | Completed | March 29, 2020
2. **Introduction to Deep Learning & Neural Networks with Keras** — IBM / Coursera | Completed | August 12, 2026
3. **Introduction to Computer Vision with TensorFlow** — Google Cloud / Coursera | Completed | August 13, 2024
4. **Python for Data Science, AI & Development** — IBM / Coursera | Completed | July 12, 2026 | Certificate
5. **AI Foundations & Industry Overview for Telecommunication** — Coursera | Completed | June 19, 2026 | Certificate
6. **Network and Security Optimization in Telecommunication** — Coursera | Completed | June 20, 2026 | Certificate
7. **4G Network Fundamentals** — Institut Mines-Télécom / Coursera | Completed | August 2, 2026 | Certificate
8. **Generative AI Engineering and Fine-Tuning Transformers** — IBM / Coursera | Completed | August 6, 2026
9. **Generative AI: Foundation Models and Platforms** — IBM / Coursera | Completed | October 1, 2024